

Submit the following five exercises. You are expected to provide full solutions in a linear and coherent fashion. You are encouraged to work in groups, but your work must be your own and must be self-contained.

Exercise 1.

1. Compute $\lim_{x \rightarrow -2} \frac{\sqrt{x^2+3} - \sqrt{x-11}}{x+2}$.
2. Compute $\lim_{x \rightarrow 3} \frac{\sqrt{x^2-x-6} - \sqrt{x-11}}{|x-3|}$.

Exercise 2.

1. Find all vertical and horizontal asymptotes of $\frac{x-4}{x^2-2x-8}$.
2. Find all vertical and horizontal asymptotes of $\frac{e^x}{e^x+2}$.

Exercise 3.

Graph a function $f(x)$ satisfying the following properties. Do not find $f(x)$ explicitly.

1. The domain of $f(x)$ is $\{x \in \mathbb{R} \mid -4 \leq x < 2\}$.
2. $f(-2) = -1$.
3. $\lim_{x \rightarrow -2} f(x) = 2$.
4. $\lim_{x \rightarrow 1^-} f(x) = 2$.
5. $\lim_{x \rightarrow 1^+} f(x) = 1$.

Exercise 4.

Graph a function $g(x)$ satisfying the following properties. Do not find $g(x)$ explicitly.

1. The domain of $g(x)$ is $\{x \in \mathbb{R}\}$.
2. $g(x)$ has a horizontal asymptote at $y = -1$.
3. $\lim_{x \rightarrow \infty} g(x) = \infty$.
4. $\lim_{x \rightarrow 2^-} g(x) = \infty$.

Exercise 5.

1. Explain what happens to e^{-x} as x approaches ∞ .
2. Explain what happens to $\sin(2x)$ as x approaches ∞ .
3. Using the above, explain what happens to $h(x) = e^{-x} \sin(2x)$ as x approaches ∞ . Suggest a value for $\lim_{x \rightarrow \infty} h(x)$.
4. Find the zeros of $h(x)$.
5. Draw a rough sketch of the graph of $h(x)$ using the information above.